

Chitragupta

Technical Reference — architecture, schema, API, security

Architecture

Channels are thin; the engine is the product. The Desk widget and the portal both POST to the same backend. No channel can bypass governance.

Layer	Modules	Responsibility
Channels	chitragupta_desk.js · portal	Chat UI, approval cards, chips, voice, upload, downloads; frappe.get_route() screen context; SSO handsha
API	app.py	FastAPI; bearer sessions; /api/sso; CORS; /api/version; /api/learning
Identity	auth.py	PBKDF2 logins; HMAC SSO verify (self-diagnosing failures); CredentialStore binds verified ERP sessions p
Engine	engine.py	Six-step loop; 12-turn conversation memory; document completion (names, dates, required fields, learned c
Language	planner.py · agent.py	Whitelisted ops (query/answer_context/explain/create/need_info/comment/report/agent/...); analytics over m
Governance	governance.py	Risk tiers (AUTO/APPROVE/BLOCK); write-gate RBAC; register_user for SSO; strict memory scope
Memory	memory.py	personal/org/skills; on_approved learning; org_norms applied to drafts; precedents; learning_stats
ERP	erp.py · erp_frappe.py	Adapter interface; FakeERP for tests; live REST (quote()/%20 paths, real error surfacing, session or token
Utilities	dates.py files.py reports.py sso.py	DD-MM-YYYY→ISO + run-on split; CSV/XLSX/PDF extraction; report generation; SQLAlchemy schema + s

Database schema

Table	Columns (essentials)	Purpose
jobs	id, actor, text, status, created	One row per user request
proposals	id, job_id, actor, doctype, draft_name, action(submit/update/cancel), value, Every staged submit any payload as doc, status, the id, leadristed	
audit	id, ts, event, actor, detail_json	Append-only trail: proposal_staged, approved(edited?), commi
memory	id, kind(personal/org), owner, scope_doctype, key, value	Preferences + learned norms (preferred_*, rate::item::supplier)
skills	name, doctype, runs, approved_clean, vetted	Per-task counters; clean_rate = the learning metric

API surface

Endpoint	Purpose
POST /api/login · /api/logout	Demo/pilot sessions (SSO replaces this in production)
POST /api/sso	{user, ts, sig, sid?, csrf?} → verified session; binds act-as-user; returns roles + acting_mode
POST /api/command	{text, context?, file?} → reply, activity[], proposals[], suggestions[], download?
POST /api/proposals/{id}/approve reject	Human decision; approve accepts edits; executes + read-back; learns
GET /api/brief · /api/memory · /api/skills · /api/audit	Daily brief; memory page (view/delete); skills; audit
GET /api/learning	clean_rate, norms, preferences, skills — the improvement scorecard
GET /api/version	Build stamp + capability flags (check FIRST when debugging)
GET /embed/chitragupta_desk.js	The widget bundle the VM pulls

Security model

- Identity: ERP-signed HMAC assertion (900s replay window, self-diagnosing failures); session binding only after the ERP confirms the sid's owner; mismatched sessions never bound
- Reads: ERPNext's permission engine is the authority (403s surfaced honestly). Writes: in-engine role gate AND ERPNext. Ledger integrity: BLOCK tier, unapprovable
- Bot token never sent on per-user calls; per-request adapter binding is thread-isolated
- Files/attachments are context, never commands; the agent's write tools only STAGE proposals; loop bounded at 8 steps
- All 131 tests include containment: jailbroken agent, malicious file, forged SSO, RBAC on every tool

Field lessons encoded (the hard-won ones)

Symptom	Root cause	Now in code
404 on PO create	Frappe 404/417 = validation too; detail was discarded	deal surfaces _server_messages
ImportError purchase+order	quote_plus in path	quote() %20 everywhere
'Please enter Req'd by Date'	item rows lacked schedule_date	_complete_doc stamps every row
'No sales in 2026' with 2026 data on screen	no date-range filtering	date_range op + apply_date_range
False 'role doesn't allow Company'	engine duplicated ERPNext permissions	reads delegated to ERPNext
Assistant re-asked known details	no conversation memory	12-turn history
Hung after ~15 requests	leaked DB sessions	scoped_session
SSO 401 with correct secret	162s VM clock drift > 120s window	900s window + self-diagnosing 401s
Ghost 'old bug' reports	deploy drift PC/VM/browser	/api/version build stamp